8	**(a)**	**(i)**	$x = 2$	A1	[1]
		(ii)	*either* beta particle *or* electron	B1	[1]
	(b)	**(i)**	mass of separate nucleons = $\{(92 \times 1.007) + (143 \times 1.009)\}$ u	C1	
			= 236.931 u	C1	
			binding energy = 236.931 u – 235.123 u		
			= 1.808 u	A1	[3]

(ii) $E = mc^2$ C1

energy $= 1.808 \times 1.66 \times 10^{-27} \times (3.0 \times 10^8)^2$

$= 2.7 \times 10^{-10}$ J C1

binding energy per nucleon $= (2.7 \times 10^{-10}) / (235 \times 1.6 \times 10^{-13})$ M1

$= 7.18$ MeV A0 [3]

(c) energy released $= (95 \times 8.09) + (139 \times 7.92) - (235 \times 7.18)$ C1

$= 1869.43 - 1687.3$

$= 182$ MeV A1 [2]

(*allow calculation using mass difference between products and reactants*)

Section B

9 **(a)** light emitting diode (*allow LED*) B1 [1]